IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART I: MEng, BEng and ACGI

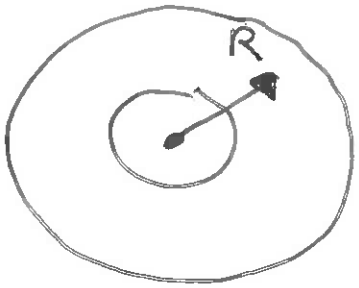
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ENERGY CONVERSION

Wednesday, 12 June 10:00 am

Time allowed: 2:00 hours

Question 2



professor drew this diagram on the board for Question 2

There are THREE questions on this paper.

Answer All questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : O. Sydoruk
Second Marker(s) : W.T. Pike

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ENERGY CONVERSION

1. a) An electric field with a strength **E** creates an electric flux Φ through an open surface S. Find the flux through this surface when the electric field strength is increased to 2**E**. [5]

b) Starting with Maxwell's equations, show that the work done by a static electric field to move a point charge between two arbitrary points does not depend on the path between the two points. [10]

c) A static electric field creates a potential of the form $\varphi(x) = e^{-x}\cos x$. Find the electric field strength. [5]

d) How does the inductance of an inductor depend on the current flowing through it? Briefly justify your answer using the definition of inductance from the formula sheet. [5]

e) Why is a single-valued permeability insufficient to characterise the magnetic properties of a ferromagnetic material? What is a better way to characterise ferromagnetic materials? [5]

f) Figure 1.1 shows an equivalent circuit of a real transformer. What two tests are used to determine the values of the components R_i, X_m, R_t, and X_t? Draw the equivalent circuit diagrams for each of these tests and state which quantities are measured. Why are these tests used instead of measuring the values of the components directly? [10]

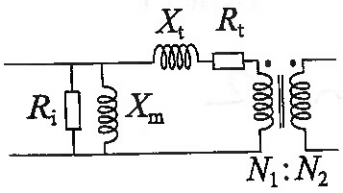


Figure 1.1

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2. An inductor is made by N turns of wire wound uniformly around a toroidal core. The core has a circular cross-section of a radius a. The radius of the toroid is R.

a) Assume that a dc current is flowing through the wire. Without doing calculations, what can be said about the magnetic field created by the current flow in the toroid? [5]

b) Derive an expression for the magnetic field strength both inside and outside of the toroid assuming that the current strength is equal to I. [15]

c) Toroidal cores may be used to build power transformers. Such transformers are attached to a metal framework using a bracket and a metal bolt as shown in Figure 2.1. The following is written in a transformer spec sheet: "Under no circumstances should both ends of the fixing bolt simultaneously contact the metal framework". Explain what may happen if this warning is ignored. [10]

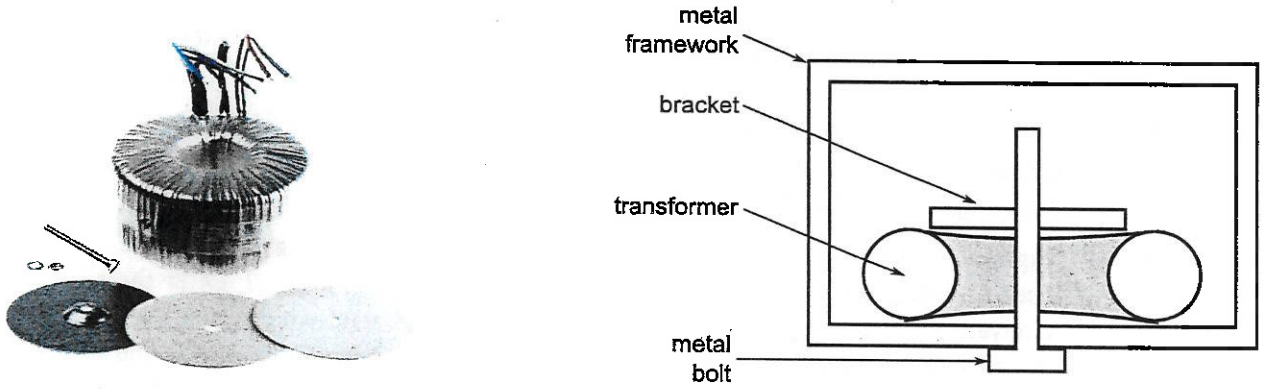
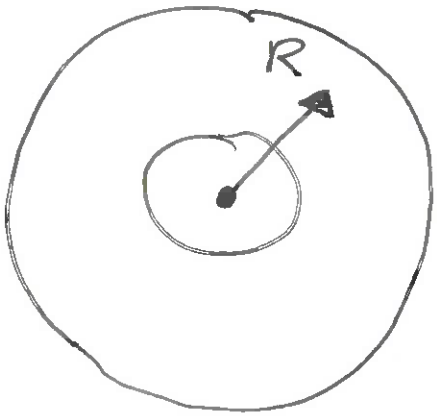


Figure 2.1

Question 2



3. A charged line segment lies on the x-axis with its centre at the origin, as shown in Figure 3.1. The segment length is $2l$. The linear charge density of the segment is $\lambda = qx/l^2$, where q is a constant.

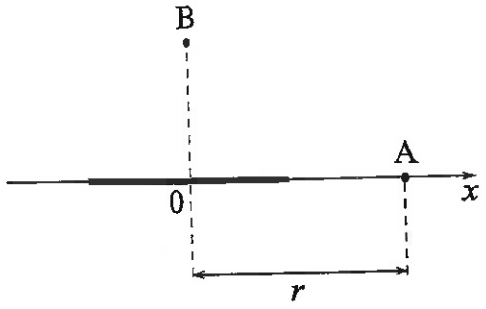


Figure 3.1

a) Find the charge of the segment. [3]

b) Using schematic drawings and without doing calculations, find the direction of the electric field at any point that lies on the x-axis to the right of the segment (i.e. when $x > l$). [3]

c) Using schematic drawings and without doing calculations, find the direction of the electric field at point B lying above the centre of the segment as shown in Figure 3.1. [4]

d) Draw schematically the electric field lines in the whole space. [5]

e) Derive an expression for the electric field strength at point A. The distance from point A to the origin is r. [15]

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